

Miles Jennings

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EDUCATION

University of California Irvine

Irvine, CA

Bachelor of Science in Computer Engineering

Sept. 2022 – June 2026

Relevant Coursework: Organization of Digital Computers, Electronics I–II, Network Analysis I–II, Data Structures and Algorithms, Discrete and Continuous-Time Signals, Digital Logic, Computer Architecture, VLSI, Embedded Systems, Digital Signal Processing, Analog IC Design

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

P2S Technology Intern

Long Beach, California

P2S Engineering

June 2025 – Present

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Supported engineers with **CAD** layout updates, wiring diagrams, and equipment schedules using AutoCAD and Revit
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs

HyperXite – SpaceX Hyperloop

Irvine, California

Control Systems Engineer

Aug. 2025 – Present

- Designed, tested, and optimized the Hyperloop pod's control system hardware using an STM32H753ZI microcontroller programmed in **C/C++**, interfacing with thermistors, time-of-flight sensors, and INA219 current sensors while evaluating components for performance and cost improvements over prior pod iterations
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI
- Implemented a **Continuous Integration (CI) pipeline** using GitHub Actions to streamline testing, version control, and code deployment across subteam

PROJECTS

AudioVisor | *C++, PDM, nRF52840*

Sept. 2025 – Present

- Built an audio-reactive visor using an nRF52840, utilizing **C++** to receive and handle digital microphone input through **PDM** communication for real-time sound magnitude/direction detection
- Prototyped and iterated hardware designs, including LED driver layouts and signal pathways, while troubleshooting performance issues throughout development
- Researched and evaluated components to satisfy sensitivity, latency, and power needs, guiding system architecture and part selection

MIPS & Pipeline Processor | *Verilog*

Jan. 2025 – Feb. 2025

- Designed and simulated a MIPS and pipelined processor using Verilog, ensuring correct instruction execution
- Implemented multiple MUX's, ALU-Controllers, and other combinational circuit components
- Developed and tested ALU controllers, multiplexers (MUXs), and other combinational logic components, optimizing data flow efficiency

Dictionary & Calculator Program | *Java, Swing*

Sept. 2024 – Dec. 2024

- Developed a dictionary application supporting import/export of text files for word and definition management
- Developed a feature-rich calculator with event-driven programming, handling factorial, logarithmic, and memory functions

Chess & Poker Program | *C, Linux, Git, SDL, GTK*

April 2024 – June 2024

- Developed a multiplayer Chess and Poker game in **C**, implementing server-to-client socket communication for real-time gameplay
- Utilized **Linux** for version control and coordination among team members.
- Designed an interactive GUI using SDL 1.2 (Chess) and GTK (Poker), enhancing user experience and engagement

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS Assembly, MATLAB

Software/Tools: Revit, AutoCAD, Bluebeam, Arduino IDE, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm